# SD-LoRA: Scalable Decoupled Low-Rank Adaptation for Class Incremental Learning

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## Background: Foundation Models + Continual Learning are Transforming AI

- Foundation models (e.g., ViTs) enable broad transfer across domains and tasks; PEFT streamlines adaptation.
- Continual Learning (CL) aims to learn sequentially without catastrophic forgetting.
- Parameter-Efficient Fine-Tuning scales adaptation without full retraining; efficiency is key.



## The Problem: Class-Incremental Learning at Scale

- Goal: recognize all seen classes without task identity at test time (no task ID).
- Challenges: catastrophic forgetting, scalability over long task sequences.
- Practical constraints: rehearsal-free training, inference efficiency, end-to-end optimization.



## Why Existing PEFT Approaches Fall Short

- Prompt/LoRA pools grow over time → selection overhead at inference.
- Some methods require storing past samples → privacy and memory issues.
- Coupled low-rank updates entangle direction and magnitude → suboptimal reuse; hurts stability–plasticity.

## Core Idea: Decouple Direction and Magnitude in LoRA

- Represent each layer update as normalized directions plus learned magnitudes to decouple direction and scale.
- Reuse early learned directions; rescale them to fit new tasks in a rehearsal-free manner.
- Train end-to-end without rehearsal; evaluate with a single final model via selection-free deployment.



## Contributions at a Glance

- SD-LoRA: rehearsal-free, inference-efficient, end-to-end CL for CIL.
- Mechanism: early directions dominate; magnitudes trace low-loss paths.
- Theory: principal components emerge under asymmetric factorization.
- Efficiency variants: SD-LoRA-RR (rank reduction) and SD-LoRA-KD (distillation) for efficiency.



## Method Overview: From Vanilla LoRA to SD-LoRA

- Vanilla LoRA: learns a new low-rank AB per task.
- SD-LoRA: fixes normalized past directions; learns magnitudes and current direction.
- Cumulative update enables direct, selection-free inference.

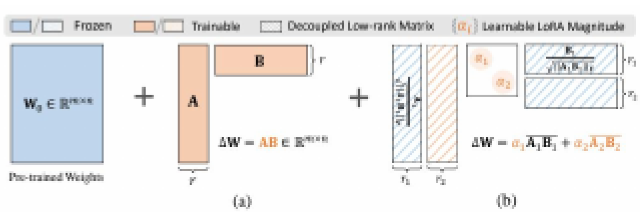


Figure: Parameter updates in (a) LoRA vs. (b) SD-LoRA at t=2; ranks $r, r_1, r_2 \ll \min\{m, n\}$.



## SD-LoRA Formulation and Training

- Layer output: $W_0 + \sum_k \alpha_k \cdot \mathrm{normalize}(A_k B_k)$ using normalized directions and $\alpha$-magnitudes.
- During task $t$: fix $W_0$ and past directions; optimize $\alpha_{1..t}$ and $A_t B_t$.
- No rehearsal data; joint optimization aligns with CL objectives (end-to-end).

## Why It Works: Shared Low-Loss Regions and Direction Reuse

- Task optima cluster near each other relative to pretrain $W_0$—this is clustering.
- Fixing an early direction and tuning its magnitude already helps.
- Early directions generalize; later ones provide fine adjustments in a low-loss region.

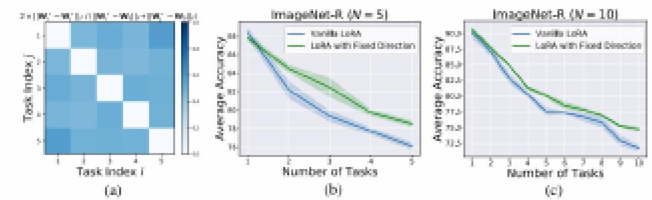


Figure: Early-direction reuse: task-specific optima remain close to $W_0$; fixing the first direction maintains accuracy across 5/10-task splits.



## Theoretical Insight: Asymmetric Matrix Factorization View

- SD-LoRA approximates factorizing a shared optimal update $\Delta W^*$ into principal components.
- Small initialization with gradient descent recovers dominant components first.
- Explains decreasing importance of later directions and rank reduction.



## Parameter-Efficient Variants

- SD-LoRA-RR: stepwise rank reduction over tasks to curb growth.
- SD-LoRA-KD: least-squares projection of new direction onto prior span; discard if residual small (distillation).
- Both maintain accuracy while improving efficiency.



## Experimental Setup: Datasets, Backbones, Metrics

- Datasets: ImageNet-R (N=5/10/20), ImageNet-A (N=10), DomainNet (N=5), CIFAR100, CUB200.
- Backbones: ViT-B/16 (supervised), DINO ViT-B/16 (self-supervised).
- Metrics: Acc (final), AAA (anytime average across tasks).

## Baselines and Protocols

- Prompt-based: L2P, DualPrompt, CODA-Prompt, HiDe-Prompt.
- LoRA-based: InfLoRA; full fine-tuning as reference.
- Standard CIL protocol: no task ID at test time; compare compute/storage.



## Results on ImageNet-R Across Task Lengths

- SD-LoRA achieves SOTA Acc and AAA across N=5/10/20.
- Largest gains at longer sequences (N=20), showing scalable performance.
- RR and KD variants remain close while reducing parameters.

Table: Performance on ImageNet-R across task lengths.

| Method | ImageNet-R (N = 5) | | ImageNet-R (N = 10) | | ImageNet-R (N = 20) | |
|---|---|---|---|---|---|---|
| | Acc↑ | AAA↑ | Acc↑ | AAA↑ | Acc↑ | AAA↑ |
| Full Fine-Tuning | 64.92(0.87) | 75.57(0.50) | 60.57(1.06) | 72.31(1.09) | 49.95(1.31) | 65.32(0.84) |
| L2P | 73.04(0.71) | 76.94(0.41) | 71.26(0.44) | 76.13(0.46) | 68.97(0.51) | 74.16(0.32) |
| DualPrompt | 69.99(0.57) | 72.24(0.41) | 68.22(0.20) | 73.81(0.39) | 65.23(0.45) | 71.30(0.16) |
| CODA-Prompt | 76.63(0.27) | 80.30(0.28) | 74.05(0.41) | 78.14(0.39) | 69.38(0.33) | 73.95(0.63) |
| HiDe-Prompt | 74.77(0.25) | 78.15(0.24) | 74.65(0.14) | 78.46(0.18) | 73.59(0.19) | 77.93(0.19) |
| InfLoRA | 76.95(0.23) | 81.81(0.14) | 74.75(0.64) | 80.67(0.55) | 69.89(0.56) | 76.68(0.57) |
| SD-LoRA | 79.15(0.20) | 83.01(0.42) | 77.34(0.35) | 82.04(0.24) | 75.26(0.37) | 80.22(0.72) |
| SD-LoRA-RR | 79.01(0.26) | 82.50(0.38) | 77.18(0.39) | 81.74(0.24) | 74.05(0.51) | 80.65(0.35) |
| SD-LoRA-KD | 78.85(0.29) | 82.47(0.58) | 77.03(0.67) | 81.52(0.26) | 74.12(0.66) | 80.11(0.75) |



## Results on ImageNet-A and DomainNet

- ImageNet-A (N=10): SD-LoRA 55.96 Acc, 64.95 AAA; +6.76 Acc and +4.03 AAA over InfLoRA (robustness).
- DomainNet (N=5): SD-LoRA 72.82 Acc, 78.89 AAA; best on both metrics (cross-domain).
- RR and KD remain competitive while preserving efficiency.



## Computation and Storage Efficiency

- Inference: 35.12 GFLOPs (same as LoRA-based), lower than prompt-pool selection.
- Params: SD-LoRA 0.37M; SD-LoRA-RR 0.23M.
- Storage: 0 stored features across methods in our family.

## Method Properties vs. Prior CL Approaches

- Only SD-LoRA is rehearsal-free, inference-efficient, and end-to-end (unified).
- No task-selection logic; one final deployed model.



## Ablation: What Design Choices Matter?

- decoupling direction and magnitude boosts Acc/AAA vs. vanilla LoRA.
- reweighting (magnitude rescaling) is critical for low-loss navigation.
- Multiple decoupled components outperform single-component decoupling.

Table: Ablation analysis of SD-LoRA. Trainable parameters are highlighted in orange.

| Training Strategy | ImageNet-R (N = 5) | | ImageNet-R (N = 10) | |
|---|---|---|---|---|
| | Acc↑ | AAA↑ | Acc↑ | AAA↑ |
| W0 + $\alpha$A1B1 | 78.17(0.27) | 81.93(0.51) | 74.82(0.96) | 80.63(0.63) |
| W0 + $\alpha$AB | 73.24(0.31) | 78.80(0.13) | 70.62(0.78) | 76.32(0.16) |
| + + $\alpha A_t B_t$ W0 + A1B1 | 78.28(0.59) | 82.02(0.71) | 74.29(0.32) | 79.74(0.71) |
| + + $\alpha_t A_t B_t$W0+ $\alpha_1$A1B1(SD-LoRA) | 79.15(0.20) | 83.01(0.42) | 77.34(0.35) | 82.04(0.24) |



## Anytime Performance with Self-Supervised Backbone

- DINO ViT-B/16: SD-LoRA maintains top AAA throughout sequences and is backbone-agnostic.
- Demonstrates consistent benefits with self-supervised features.
- Consistent lead across ImageNet-A/R setups.

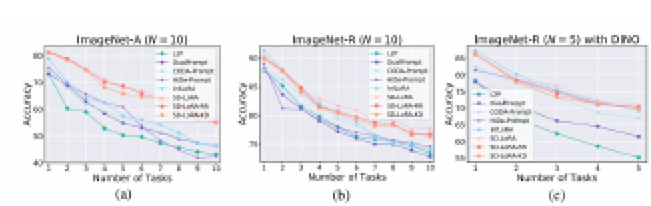


Figure: Average accuracy during sequential training on ImageNet-A/R using DINO ViT-B/16.



## Low-Loss Trajectory: Avoiding Forgetting While Improving

- Vanilla LoRA trades T1 for T2 along update paths.
- SD-LoRA scales prior directions → overlapping low-loss region.
- Improves new tasks without degrading past ones (stability–plasticity).

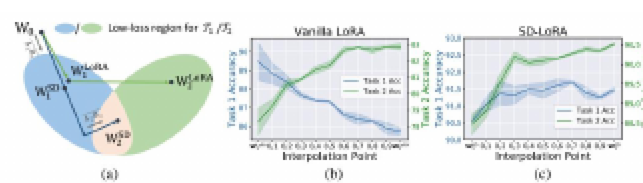


Figure: Two-task training trajectories and accuracy: vanilla LoRA forgets; SD-LoRA reaches an overlapping low-loss region.

## How New Directions Emerge and Are Weighted

- New directions initially align with the prior span; residual grows slowly, exhibiting alignment.
- Magnitudes emphasize early directions; later ones diminish, showing magnitude decay.
- Supports reuse-first, refine-later learning dynamics.

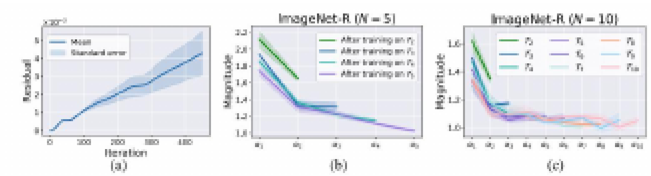


Figure: Learning analysis of SD-LoRA: (a) least-squares residual of new vs. prior directions; (b)(c) magnitudes $\alpha_k$ across 5/10-task ImageNet-R sequences.



## Extended Horizons: $\alpha$ Magnitudes Over Many Tasks

- Systematic decay of $\alpha$ for later directions on long sequences, demonstrating scalability.
- Early-task components dominate across domains.
- Corroborates RR/KD efficiency design choices.

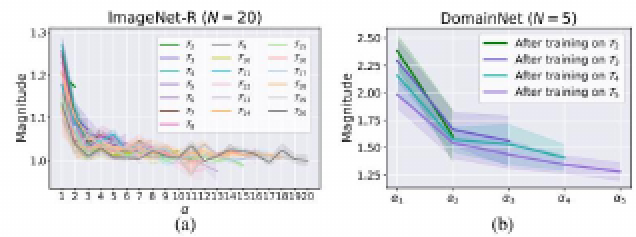


Figure: LoRA magnitude trends $M = \{\alpha_k\}_{k=1}^{N}$ over long horizons: ImageNet-R (N=20) and DomainNet (N=5).



## Clustering Holds Across Backbones and Datasets

- Task-specific optima cluster closer to each other than to $W_0$—evidence of clustering.
- Observation consistent on DomainNet and ImageNet-R with DINO.
- Supports the shared low-loss region hypothesis.

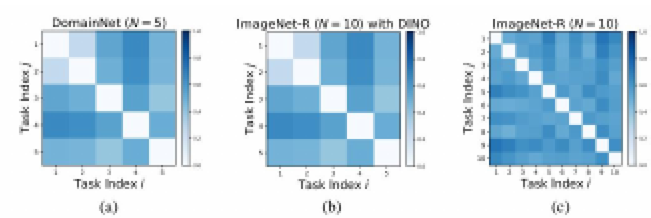


Figure: Relative distances among task optima vs. $W_0$ on DomainNet (N=5) and ImageNet-R (N=5,10) with DINO ViT-B/16.



## Limitations and Practical Considerations

- Assumes shared low-loss region; extreme domain shifts may reduce reuse (limitations).
- Hyperparameters for RR/KD (ranks, residual $\tau$) require tuning.
- Future: extend beyond ViTs; integrate with other PEFT modules (hybrid PEFT).

## Conclusions: What SD-LoRA Delivers

- rehearsal-free, selection-free, end-to-end CIL with a single deployed model.
- Decoupling direction/magnitude yields low-loss trajectories and strong stability–plasticity.
- RR and KD variants enhance parameter efficiency with minimal performance loss.



## Future Work & Impact

- Backbones: CNNs, larger ViTs, multimodal foundation models.
- hybrid PEFT: combine with adapters/prefixes for complementary gains.
- Theory-guided rank/distillation schedules for higher efficiency.



## Questions & Discussion

- Thank you for your attention! Questions and feedback are welcome.



## Acknowledgments

- We thank collaborators, lab members, and funding agencies for their support.
- Code available: `https://github.com/WuYichen-97/SD-Lora-CL`